

Mihai Sorin Dobre

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PROFILE

AI technology leader, researcher and hands-on engineer with a PhD in AI and 10+ years of experience spanning applied AI research, production ML and startup technology leadership. Currently Co-founder & CTO of Klaris AI; previously Lead Research Scientist at Five AI/Bosch. Experienced in building and leading research and engineering teams and taking AI systems from research through production, with expertise in agentic AI, LLMs, reinforcement learning, multi-agent systems and probabilistic modelling.

Technical expertise: Agentic AI & LLMs · RAG & retrieval · Deep Learning · Reinforcement Learning · Multi-Agent Systems · Bayesian Methods · Prediction & Planning

PROFESSIONAL EXPERIENCE

Co-founder & CTO
Klaris AI, UK

Dec 2024 – present
klaris.ai

- Co-founded Klaris AI, an agentic AI platform for MedTech regulatory compliance; raised \$1M and built the technology function from zero, spanning AI/ML, software engineering, cloud infrastructure, security and technical hiring
- Architected and developed the core AI platform, combining custom retrieval, knowledge graphs, agentic workflows and OCR to analyse heterogeneous regulatory documentation against expert-validated requirements
- Established the company's AI evaluation and research programme, benchmarking frontier models and methods against public and proprietary datasets; achieved >89% retrieval recall, >95% finding accuracy and <2% hallucination rate, vs at most 15% requirement coverage from leading off-the-shelf models (e.g. OpenAI, Claude Opus, Codex)
- Built the production technology platform from prototype to customer deployment, initially owning backend, database and infrastructure directly while coordinating frontend engineering; platform subsequently used across dozens of customers and pilots
- Hired and lead the engineering team, owning technical strategy and product roadmap across research, engineering and customer priorities
- Established security, software development, CI/CD and documentation processes and led the company through ISO/IEC 27001:2022 certification and subsequent audits

Entrepreneur in Residence
Antler, London, UK

Sep 2024 – Dec 2024

- Completed the Antler incubator programme and secured funding to found Klaris AI
- Partnered with co-founder Francesco Corazza to validate the MedTech regulatory compliance opportunity

Lead Research Scientist
Five AI / Bosch, UK

May 2023 – Jul 2024

- Led the Motion Planning and Prediction applied research team: technical direction, cross-org collaboration with Bosch and academia, and line management
- Set research direction for multi-agent prediction and planning, including robustness under distribution shift and tighter integration of learned prediction with planning
- Developed hybrid approaches combining GNNs, Transformers, multi-agent RL and imitation learning with classical planning and Bayesian inference

- Delivered ML solutions for HD Map Reconstruction from camera and LiDAR for autolabelling; guided engineering to bring research into the deployed vehicle stack
- Extended the research simulator to multi-agent settings and accelerated it with JAX (vectorisation, JIT, hardware acceleration)

Senior Research Scientist in Motion Planning and Prediction
Five AI, UK

May 2020 – May 2023

- Led Interactive Prediction and Planning: MCTS ego planning with sampled predictions, and multi-agent models of simultaneous interaction
- Built prediction via neural models with Bayesian goal recognition; occlusion handling via optimisation-based planning and belief modelling (e.g. particle filtering)
- Developed evaluation frameworks and metrics for prediction and planning to guide product priorities

Research Scientist in Motion Planning and Prediction
Five AI, UK

May 2018 – May 2020

- Designed a high-level planner for complex manoeuvres (e.g. roundabout merging, overtaking) and a research driving simulator used across RL, interactive planning and prediction work

Software Developer on STAC project
University of Edinburgh, UK

Sep 2014 – Mar 2015 & Aug – Dec 2016

- Collaborated with Toulouse and Heriot-Watt on a rule-based Settlers of Catan agent and ML methods for negotiation

Software Developer Intern on WebSphere Application Server
IBM, Hursley, UK

Jul 2011 – Aug 2012

SELECTED PUBLICATIONS

- Aron D, Yiwei W, Faris J, Marcel H, Mihai D, Alexander L, Joschka B, and Johannes B. Driving is a game: Combining planning and prediction with bayesian iterative best response. In *ECCV*, Malmö, Sweden, 2026. to appear
- Morris A*, Mihai D*, Stefano A, John R, and Subramanian R. Flash: Fast and light motion prediction for autonomous driving with bayesian inverse planning and learned motion profiles. In *IEEE IROS*, Kyoto, Japan, 2022
- Josiah H, Arrasy R, Elliot F, Francisco E, Mihai D, John R, Subramanian R, and Stefano A. Interpretable goal recognition in the presence of occluded factors for autonomous vehicles. In *IEEE IROS*, Prague, Czech Republic, 2021
- Stefano A, Cillian B, John W, Balint G, Francisco E, Mihai D, and Subramanian R. Interpretable goal-based prediction and planning for autonomous driving. In *IEEE ICRA*, Xi'an, China, 2021
- Mihai Dobre and Alex Lascarides. POMCP with human preferences in settlers of catan. In *Proceedings of the Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, Edmonton, Canada, 2018
- Joachim F, Ben K, Mihai D, Marco D, Emmanuel K, Daniel D, Bonnie W, and Federico F. Talking to myself: self-dialogues as data for conversational agents. *CoRR*, abs/1809.06641, 2018
- Ben K, Marco D, Mihai D, Daniel D, Joachim F, Federico F, Emmanuel K, Jianpeng C, and Bonnie L. W. Edina: Building an open domain socialbot with self-dialogues. *CoRR*, abs/1709.09816, 2017

EDUCATION

PhD, Informatics

2013 – 2019

Low-resource learning in complex games

University of Edinburgh, UK

Supervisors: Alex Lascarides, Subramanian Ramamoorthy

BSc (Hons) Robotics with Artificial Intelligence (1st class)

2009 – 2013

University of Bradford, UK

TECHNICAL EXPERTISE

AI/ML: LLMs, agentic systems, RAG/retrieval, PyTorch, JAX, TensorFlow, RLlib, deep learning, reinforcement learning, probabilistic modelling

Software: Python, C++, SQL, Java

Platform: AWS, Azure, Docker, CI/CD, Git, Linux

AWARDS

- Best Student Paper Award for “Exploiting action categories in learning complex games”
- Best Overall Performance Award received for BSc degree (2013)

PREVIOUS PROJECTS AND EXPERIENCE

- *Amazon Alexa Challenge 2017:* socialbot for open-domain conversation with team Edina (University of Edinburgh).
- *Strategic Conversation (STAC):* models combining linguistic theory, agent interaction and decision making.
- Teaching assistant at Edinburgh (Reasoning and Agents; Reinforcement Learning) and Bradford (Software Development), including lecturing, labs and assessment for large cohorts.

OTHER INTERESTS

Tennis: competed on the ITF World Pro Circuits, on Romanian and British Tour, and University team levels.